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Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: seaborn in /usr/local/lib/python3.12/dist-packages (0.13.2)
Requirement already satisfied: plotly in /usr/local/lib/python3.12/dist-packages (5.24.1)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.4.9)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (25.0)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.2.5)
Requirement already satisfied: tenacity>=6.2.0 in /usr/local/lib/python3.12/dist-packages (from plotly) (8.5.0)
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.16.3)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.5.2)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
✓ All packages imported successfully!
```

 GROCERY BASKET SIMILARITY ANALYZER

Choose your data input method:

- 1. Upload CSV file
- 2. Use sample data
- 3. Create custom baskets manually

Enter your choice (1/2/3): 2

✓ Loading sample data...
Shape: 7 baskets × 7 items

 DATA OVERVIEW

1. Basket-Item Matrix:

	milk	bread	eggs	butter	cheese	yogurt	jam
Customer A	2	1	1	1	1	0	0
Customer B	1	2	1	0	0	2	0
Customer C	2	0	2	1	2	1	0
Customer D	0	3	0	1	0	0	2
Customer E	1	0	0	1	1	3	0
Customer F	3	2	2	2	1	1	0
Customer G	0	1	0	0	0	0	3

2. Summary Statistics:

	milk	bread	eggs	butter	cheese	yogurt	jam
count	7.000000	7.000000	7.000000	7.000000	7.000000	7.000000	7.000000
mean	1.285714	1.285714	0.857143	0.857143	0.714286	1.000000	0.714286
std	1.112697	1.112697	0.899735	0.690066	0.755929	1.154701	1.253566
min	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
25%	0.500000	0.500000	0.000000	0.500000	0.000000	0.000000	0.000000
50%	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	0.000000
75%	2.000000	2.000000	1.500000	1.000000	1.000000	1.500000	1.000000
max	3.000000	3.000000	2.000000	2.000000	2.000000	3.000000	3.000000

3. Total items per basket:

	0
Customer F	11
Customer C	8
Customer A	6

Customer B 6
Customer D 6
Customer E 6
Customer G 4

dtype: int64

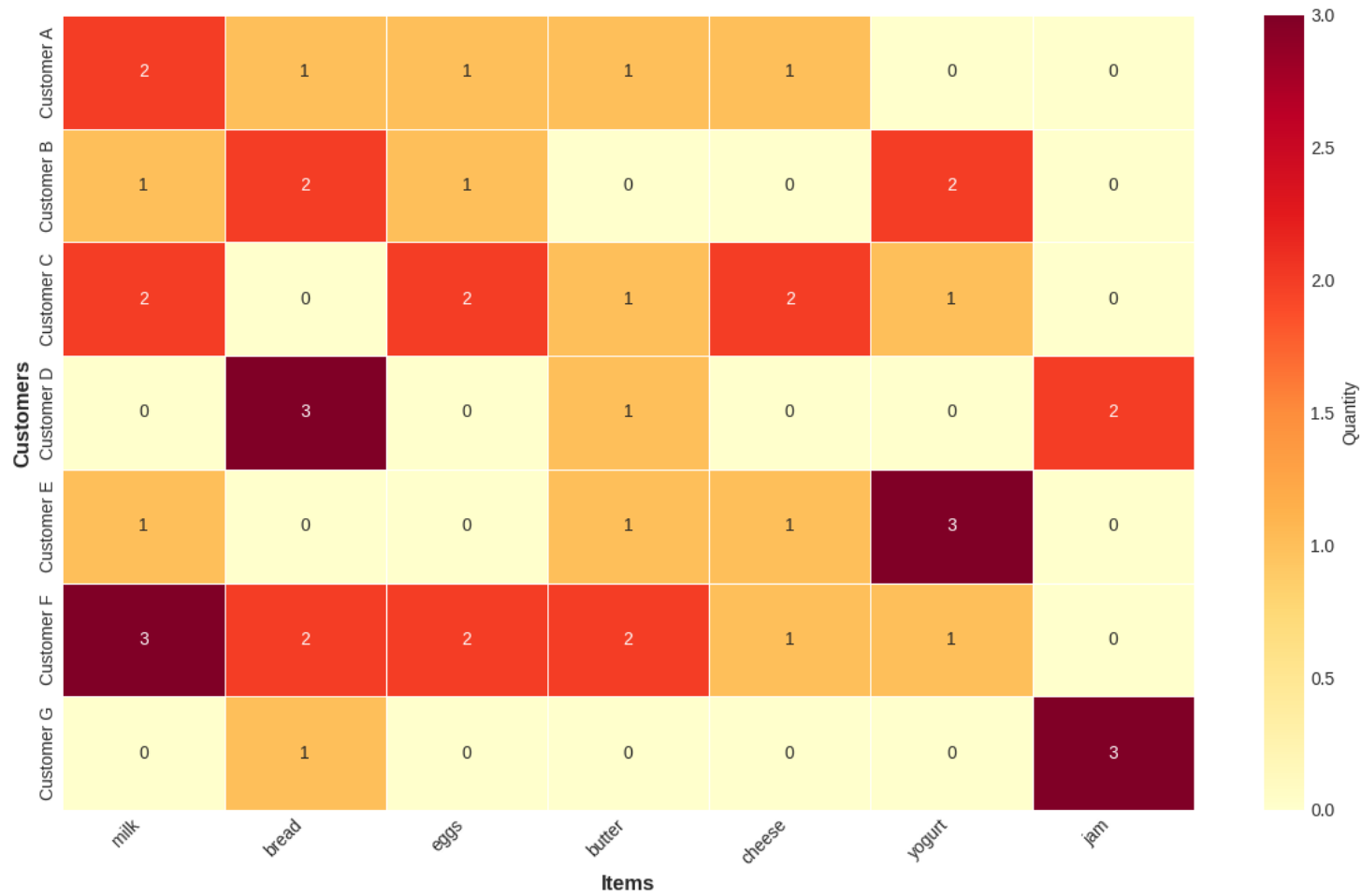
4. Item popularity (total quantity across all baskets):

	0
milk	9
bread	9
yogurt	7
eggs	6
butter	6
cheese	5
jam	5

dtype: int64

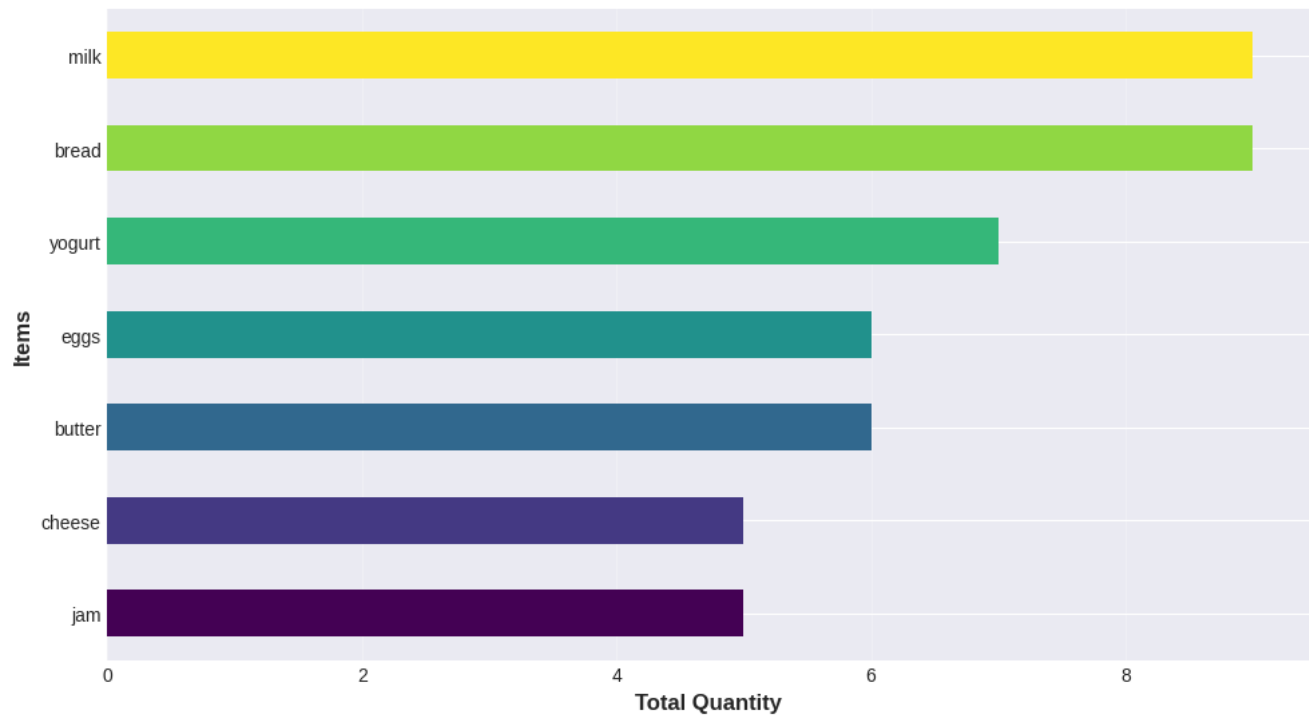
GENERATING VISUALIZATIONS

Grocery Basket Heatmap
(Quantities of Items per Customer)



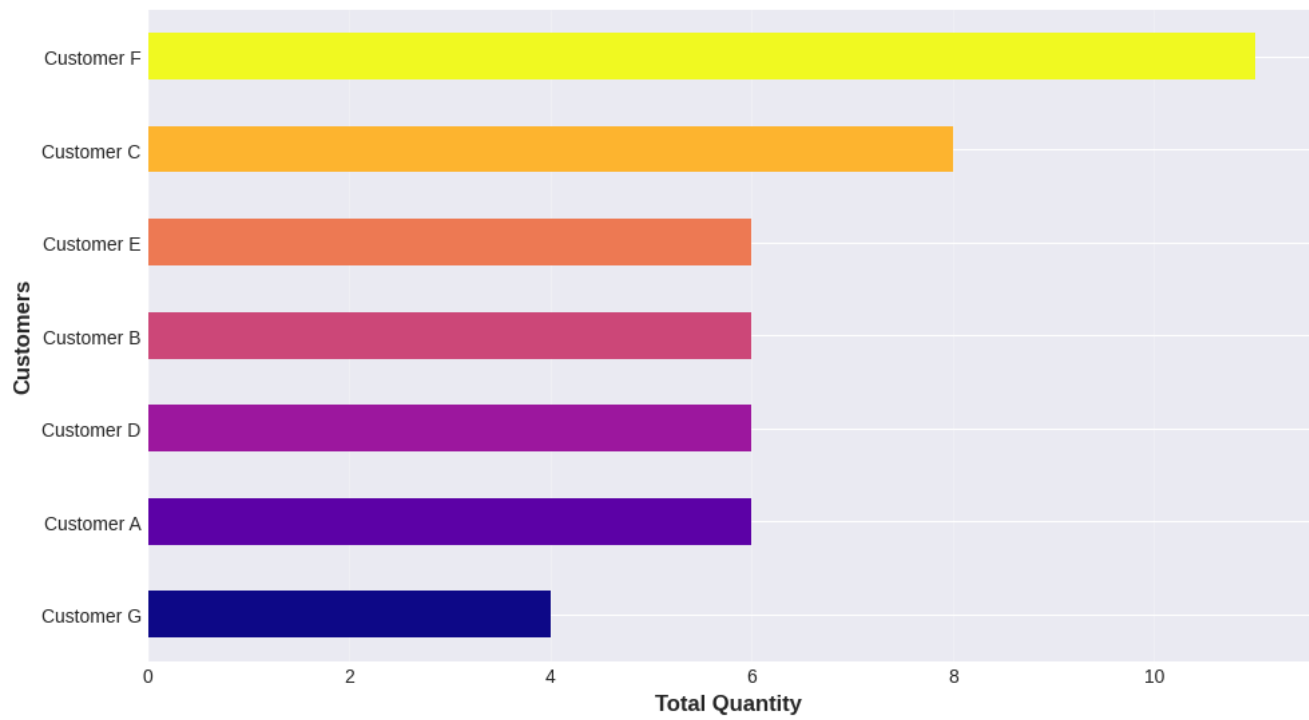
Heatmap generated!

Item Popularity Across All Baskets



✔ Bar chart generated!

Total Items per Basket



✔ Basket comparison generated!

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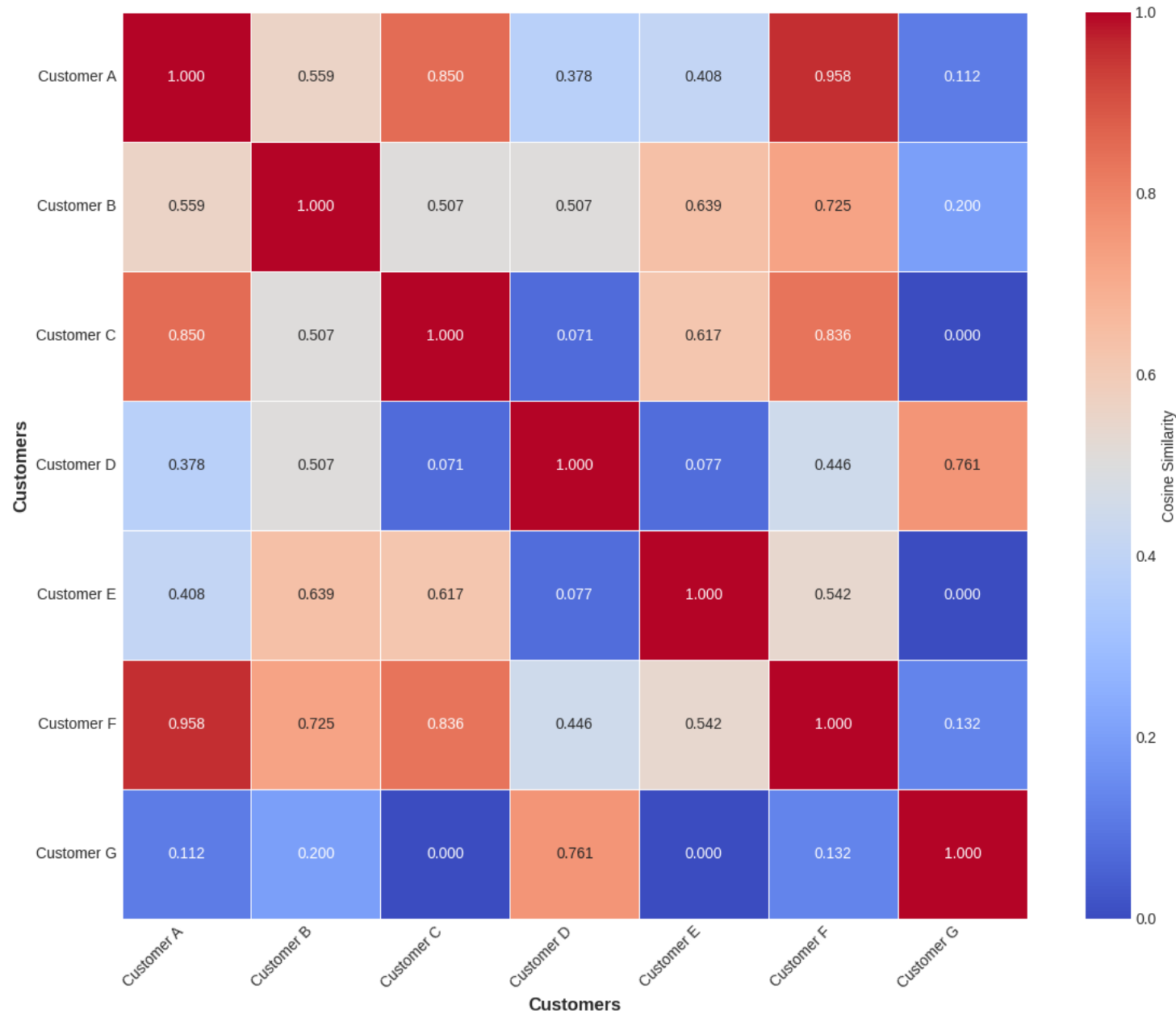
🧮 CALCULATING COSINE SIMILARITY

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✔ Similarity Matrix:

	Customer A	Customer B	Customer C	Customer D	Customer E	Customer F	Customer G
Customer A	1.000	0.559	0.850	0.378	0.408	0.958	0.112
Customer B	0.559	1.000	0.507	0.507	0.639	0.725	0.200
Customer C	0.850	0.507	1.000	0.071	0.617	0.836	0.000
Customer D	0.378	0.507	0.071	1.000	0.077	0.446	0.761
Customer E	0.408	0.639	0.617	0.077	1.000	0.542	0.000
Customer F	0.958	0.725	0.836	0.446	0.542	1.000	0.132
Customer G	0.112	0.200	0.000	0.761	0.000	0.132	1.000

Cosine Similarity Matrix
(1.0 = Identical, 0.0 = No Similarity)



✔ Similarity matrix visualization generated!

🎯 BASKET SIMILARITY ANALYSIS

Available baskets:

- 1. Customer A
- 2. Customer B
- 3. Customer C
- 4. Customer D

- 5. Customer E
- 6. Customer F
- 7. Customer G

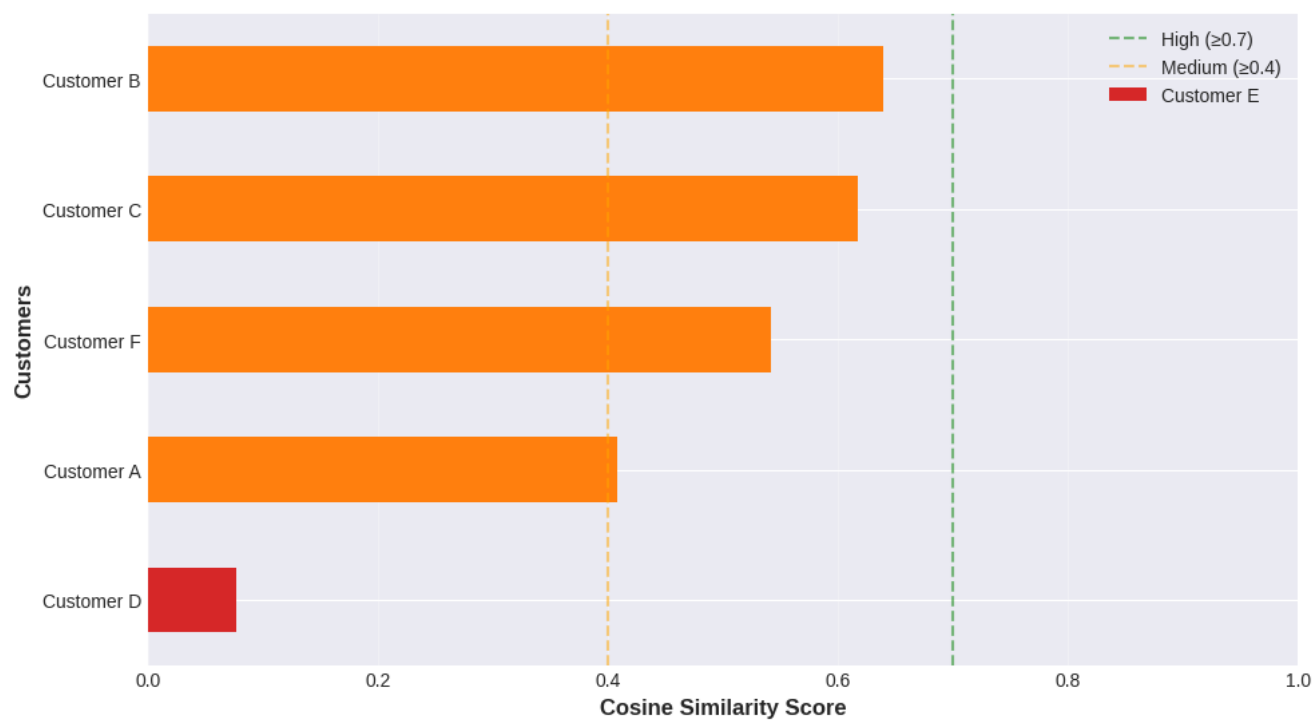
Select a basket number to analyze: 5

🇮🇹 Analyzing similarities for: Customer E
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🏆 Top Similar Baskets:			

1. Customer B	🟡 MEDIUM		63.90%
Customer E: [1 0 0 1 1 3 0]			
Customer B: [1 2 1 0 0 2 0]			
2. Customer C	🟡 MEDIUM		61.72%
Customer E: [1 0 0 1 1 3 0]			
Customer C: [2 0 2 1 2 1 0]			
3. Customer F	🟡 MEDIUM		54.17%
Customer E: [1 0 0 1 1 3 0]			
Customer F: [3 2 2 2 1 1 0]			
4. Customer A	🟡 MEDIUM		40.82%
Customer E: [1 0 0 1 1 3 0]			
Customer A: [2 1 1 1 1 0 0]			
5. Customer D	🔴 LOW		7.72%
Customer E: [1 0 0 1 1 3 0]			
Customer D: [0 3 0 1 0 0 2]			

Top 5 Similar Baskets to "Customer E"



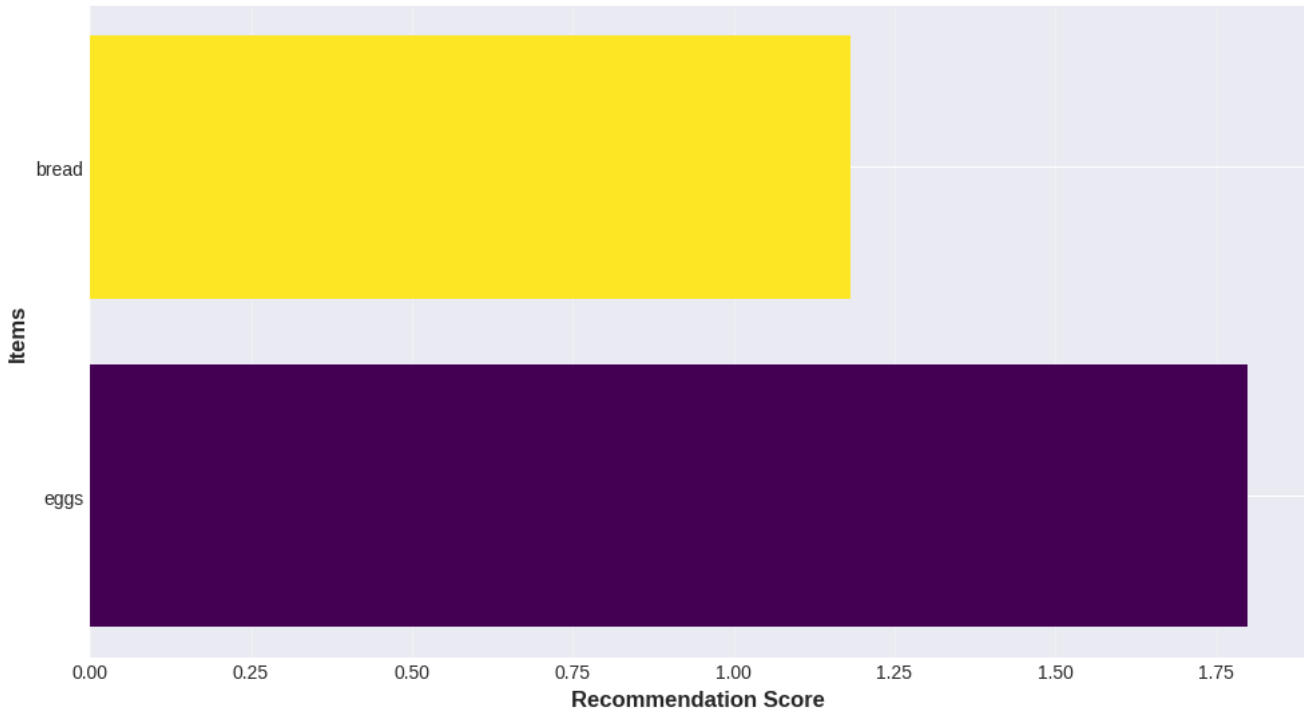
✅ Similarity chart generated!

💡 SMART PRODUCT RECOMMENDATIONS
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🛍 Recommended items for Customer E:			

1. eggs	Confidence: 100.0%	Score: 1.798	
2. bread	Confidence: 65.7%	Score: 1.181	

Top Recommended Items for "Customer E"



✓ Recommendations chart generated!

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🎨 GENERATING INTERACTIVE VISUALIZATIONS

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3D Basket Similarity Visualization (PCA)



✓ Interactive 3D plot generated!

Basket Similarity Network (Threshold > 0.3)



✔ Interactive network plot generated!

📄 ANALYSIS SUMMARY REPORT

🔄 OVERALL STATISTICS

- Total Baskets Analyzed: 7
- Total Unique Items: 7
- Average Basket Size: 6.71 items
- Most Popular Item: milk (9 total)
- Least Popular Item: cheese (5 total)

🔄 SIMILARITY INSIGHTS FOR "Customer E"

- Most Similar Basket: Customer B (Score: 0.639)
- Least Similar Basket: Customer G (Score: 0.000)
- Average Similarity: 0.381
- Items in Basket: 6

🔄 ALGORITHM DETAILS

- Method: Cosine Similarity
- Formula: $\cos(\theta) = \frac{A \cdot B}{(|A| \times |B|)}$
- Range: 0.0 (no similarity) to 1.0 (identical)
- Vector Dimensions: 7

🔄 RECOMMENDATIONS

- Total Recommendations Generated: 2
- Top Recommended Item: eggs
- Recommendation Confidence: 100.0%

✔ ANALYSIS COMPLETE!

- 📄 Do you want to export results? (y/n): y
- ✔ Similarity matrix exported to 'similarity_matrix.csv'
 - ✔ Recommendations exported to 'recommendations_Customer_E.csv'

🎉 Thank you for using Grocery Basket Similarity Analyzer!